Homework 5 due April 23rd

Exam #2: April 28th, 8:30 AM

Heat equation in 2D

$$u_t = u_{xx} + u_{yy}$$

$$u(x,y,t)$$

$$u(x,y,t=0) = \eta(x,y)$$

$$(x,y) \in [0,1] \times [0,1] = \Omega$$

$$u(x,y,t) = g(x,y,t) \text{ for } (x,y) \in \partial\Omega$$

$$x_i = ih$$
$$y_j = jh \quad i,j = 0,1,\ldots,m+1$$
$$h = \frac{1}{m+1}$$

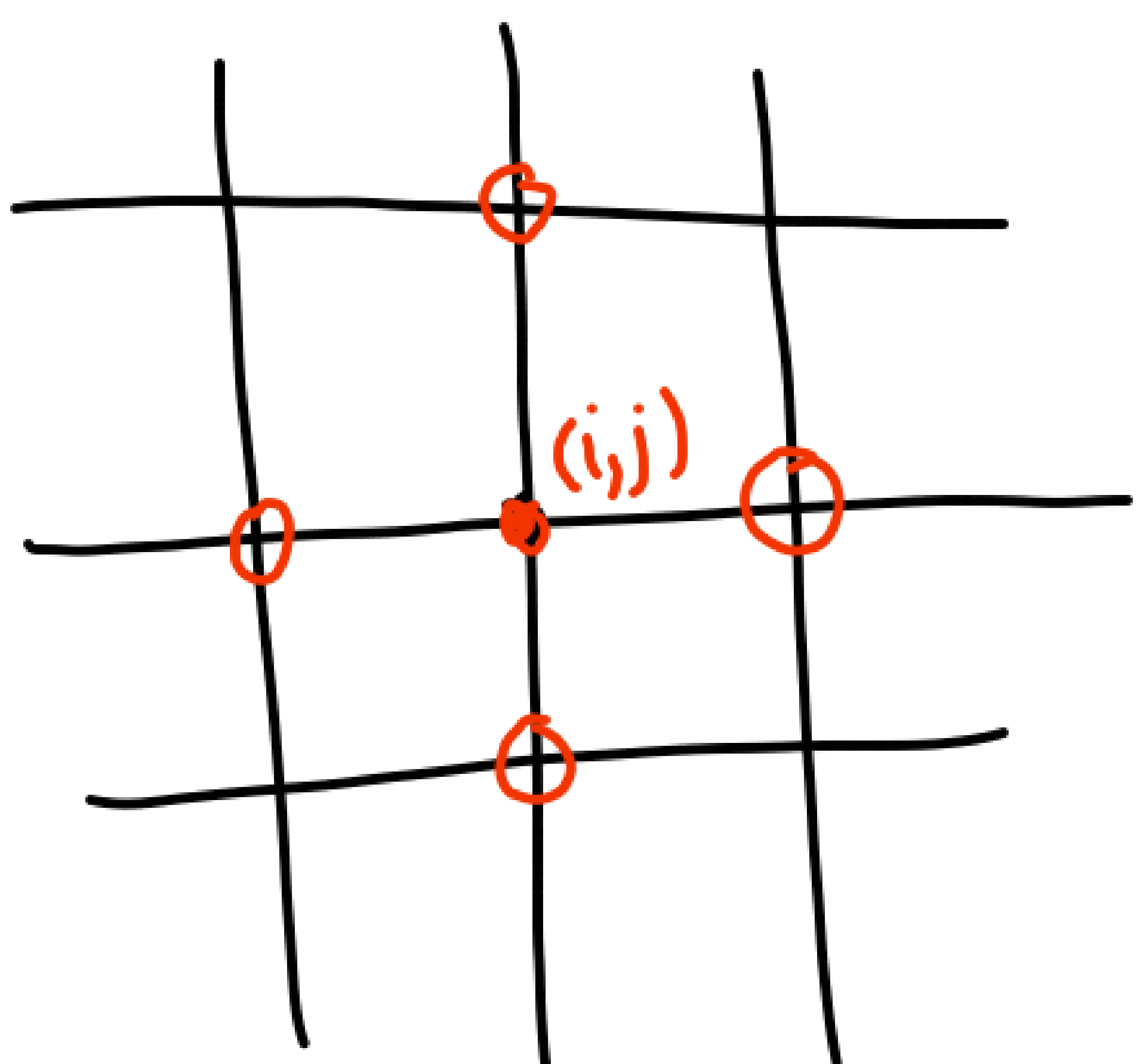
$$U_{ij}(t) \approx u(x_i, y_j, t)$$

$$U_{ij}^n \approx u(x_i, y_j, t_n)$$

$$U_{ij}^n = \begin{bmatrix} U_{11} \\ U_{21} \\ \vdots \\ U_{m1} \\ U_{12} \\ \vdots \\ U_{mm} \end{bmatrix}$$

$$u_{xx}\big|_{(x_i,y_j,t_n)} \approx \frac{U^n_{i+1,j}-2U^n_{ij}+U^n_{i-1,j}}{h^2} = (D_x^2 U)_{ij}$$

$$u_{yy}\big|_{(x_i,y_j,t_n)} \approx \frac{U^n_{i,j+1}-2U^n_{ij}+U^n_{i,j-1}}{h^2} = (D_y^2 U)_{ij}$$



5-point Laplacian

$$U'(t) = \nabla_h^2 U(t)$$

system of m^2 ODEs.

$$\nabla_h^2 = D_x^2 + D_y^2$$

Implicit trapezoidal method (Crank–Nicolson)

$$U^{n+1} = U^n + \frac{k}{2}\left(\nabla_h^2 U^n + \nabla_h^2 U^{n+1}\right)$$

$$\underline{\left(I - \frac{k}{2}\nabla_h^2\right)} U^{n+1} = \left(I + \frac{k}{2}\nabla_h^2\right) U^n$$

System of linear algebraic equations (need to solve at each step)

$$\nabla_h^2 = \frac{1}{h^2}$$

What solver should we use?

- Direct (LU factorization)
  - + Reuse factors at each step
  - $\mathcal{O}(m^4)$ operations

- Iterative methods (CG, multigrid)
  - Efficiency depends on:
    1. initial guess
    2. Condition #

$$K\left(I - \frac{k}{2}\nabla_h^2\right) = \frac{\max|\lambda|}{\min|\lambda|} \approx \frac{\mathcal{O}(1/h)}{\mathcal{O}(1)} = \mathcal{O}(1/h)$$

Often, methods converge in 1-2 iterations

→ can be more efficient than direct methods

For larger problems, iterative methods are the only viable approach because of memory considerations.

Dimensional Splitting

Instead of solving

$$u_t = u_{xx} + u_{yy},$$

let's alternate solving

$$u_t = u_{xx}$$

and

$$u_t = u_{yy}.$$

$$\left(I - \frac{k}{2} D_x^2\right) U^* = \left(I + \frac{k}{2} D_x^2\right) U^n \quad (1)$$

$$\left(I - \frac{k}{2} D_y^2\right) U^{n+1} = \left(I + \frac{k}{2} D_y^2\right) U^* \quad (2)$$

Solve (1) for U^*
then solve (2) for U^{n+1}.

We need to solve $2m$ systems of size m (tridiagonal!)

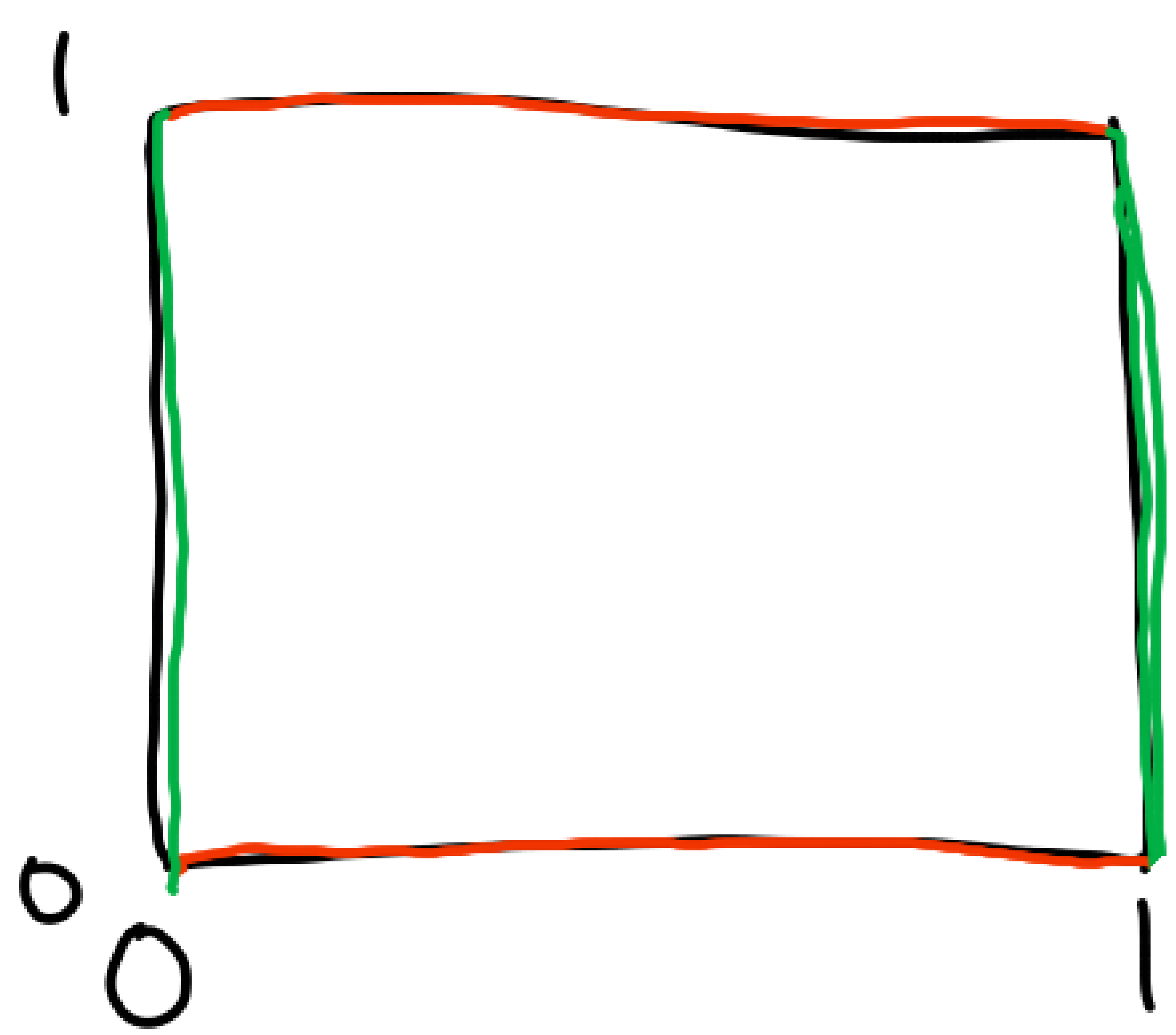
LU cost: $\mathcal{O}(m^3)$

Only 1st-order accurate

Boundary Conditions for U^*

In (2): Need BCs at top and bottom

Take BCs at $t = t_n$ and diffuse in x (solve (1) for the top and bottom rows)



$$\sum_j \alpha_j U^{n+j} = k \sum_j \beta_j f(U^{n+j})$$

$$u'(t) = \lambda u \qquad f(u) \to \lambda u$$

$$\sum_j \alpha_j U^{n+j} = z \sum_j \beta_j U^{n+j}$$

$$U^n \to \zeta^n$$

In this case:

$$(3 - \underline{2z})\zeta^2 - 4\zeta + 1 = 0$$

Do all roots $\to 0$ as $z \to -\infty$?

as $z \to -\infty$, we have approximated

$$-2z\zeta^2 = 0$$

$$\zeta^2 = 0 \qquad \zeta_1 = \zeta_2 = 0$$

$$\sum_j (\alpha_j - z\beta_j)\zeta^j = \pi(\zeta; z) = \rho(\zeta) - z\sigma(\zeta)$$